	If the load (R_l) in the common emitter amplifier is increased, then the voltage gain w decrease
	Ideal op-amp has high gain and low input resistance
	The zener diode is open circuit when the magnitude of reverse bias voltage is less than zener voltage V_Z
4	The gain of the standard non-inverting op-amp may be 0.75
5	When the input to the integrator is triangle wave, then the output is saw tooth
6	For the negative clipper circuit, the positive peaks are not clipped because the diode forward bias
7	When the input to the integrator is sine wave , then the output is out- of phase with the in signal by 180°
8	The capacitor in the integrator op-amp is in the feedback terminal
9	The rectifier circuits used for converting AC to pulsating DC
10	Filters used to maximize the ripple of rectifier output
11	The common –collector amplifier has voltage gain less than 1
12	The output ripple frequency of the full-wave rectifier is the same of the input signal
13	The output signal of a common –collector amp. is out of phase with 180°
14	Half-wave is an example of a limiter circuit
15	The common –base amp. Has a low input resistance
16	If the bypass capacitor in a CE amp. is removed , then the voltage gain will increase
17	Current gain in a transistor is inversely proportional to collector current
18	With a common-collector (C-C) amplifier the input is applied to the base and the outp taken from the emitter.
19	A decrease in base current of a C-E amplifier causes the voltage measured betv emitter and the collector to increase
20	If the load (R_L) in the common-source amplifier is increased, then voltage gain will decreas

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In an inverting Op-Amp circuit, answer the following questions, if R_f = 10kΩ and R_i = 1k

13. The voltage gain is:
(a) 1 (b) 11 (c) 10 (d) 0.1

14. The output signal out-of-phase with the input by
(a) 0° (b) 45° (c) 90° (d) 180°

15. The maximum peak-to-peak input voltage to get undistorted output, if the Op-Amp biased with +15V and -15V dc.
(a) 1.5 (b) 3 (c) 15 (d) 30

In a non-inverting Op-Amp circuit, answer the following questions, if R_f = 10kΩ and R 1kΩ

16. The voltage gain is:
(a) 1 (b) 11 (c) 10 (d) 0.1

17. The output signal out-of-phase with the input by
(a) 0° (b) 45° (c) 90° (d) 180°

18. The maximum value of R_f to get undistorted output, if the Op-Amp is biased with +1 and -15V dc and the input is 2 V_{p-p}.
(a) 14kΩ (b) 30kΩ (c) 15kΩ (d) 16kΩ

In an Op-Amp differentiator circuit, answer the following questions, if R_f = 22kΩ, R 2.2kΩ, and C = 4.7nF.

19. The output signal out-of-phase with the input by
(a) 0° (b) 45° (c) 90° (d) 180°

20. When the input is a triangle wave, the output of the *differentiator* is a
(a) square wave (b) sine wave (c) triangle wave (d) Saw tooth

21. The low-frequency response is approximately:
(a) 15.4KHz (b) 154KHz (c) 2.45KHz (d) 24.5KHz

22. The output peak voltage, when the input is a sin wave with 1V_p, 20KHz frequency:
(a) 1 (b) 11 (c) 10 (d) 0.1

In an Op-Amp integrator circuit, answer the following questions, if R_f = 100kΩ, R_i = 2.2 nF.

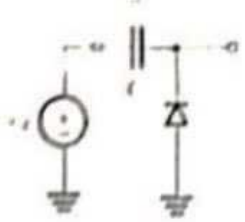
23. The output signal out-of-phase with the input by
(a) 0° (b) 45° (c) 90° (d) 180°

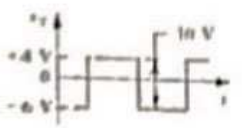
24. When the input is a square wave, the output of the oscillator is a
(a) square wave (b) sine wave (c) triangle wave (d) Saw tooth

25. The low-frequency response is approximately:
(a) 11.5KHz (b) 1.15KHz (c) 7.2KHz (d) 0.72KHz

26. The output peak voltage, when the input is a sin wave with 1V_p, 2KHz frequency:
(a) 1 (b) 3.61 (c) 10 (d) 0.1

27) For the circuit and the input signal shown which of the following is true for the outp voltage?

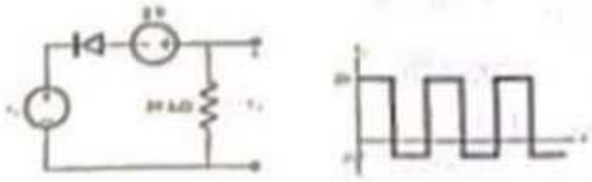


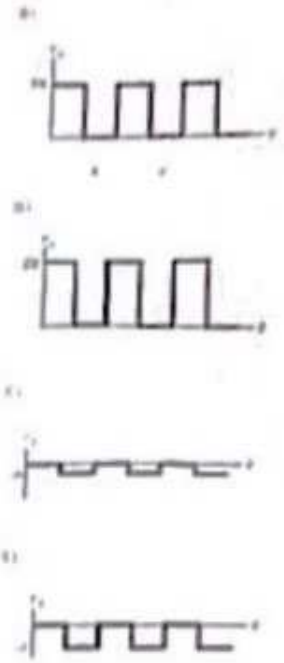


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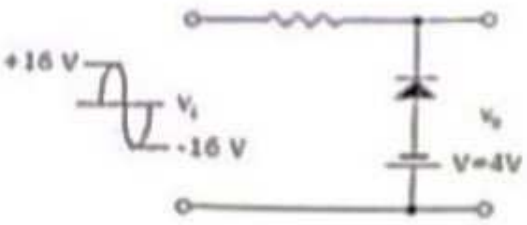
a) The output waveform is a square wave with lowest peak clamped to 0V
b) The output wave is a square wave with lowest peak clamped to -6V
c) The output waveform is a square wave with highest peak clamped to 4V
d) The output waveform is a straight line with the value of output voltage equal to 10

28)For the given circuits and input waveform determine the output waveform





29) Determine the peak for both half cycles of the output waveform.



a) - 16 V, -4 V b) 16V, 4V c) -16V ,4 V d) 16V ,-4 V

30)

When a Zener diode circuit is used to stabilize the output voltage given a fixed load resistor and a variable input voltage, the input voltage must be:

(a) small enough to turn-off the Zener diode
(b) large enough to turn-off the Zener diode
(c) small enough to turn-on the Zener diode
(d) large enough to turn-on the Zener diode
(e) none of the above

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